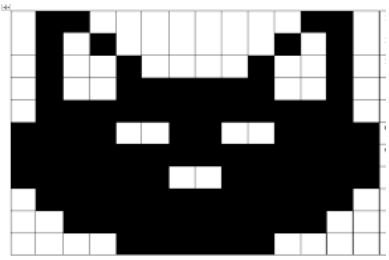
Unit III: Generalized linear models

Chapter 18 highlights

- Comparisons of proportions or odds

What do these images have in common?



Binary response for "yes" "no" data

How can we map two responses to two numbers?

What summaries can we use to summarize the response? Compare the response across groups?

Parameter Definitions

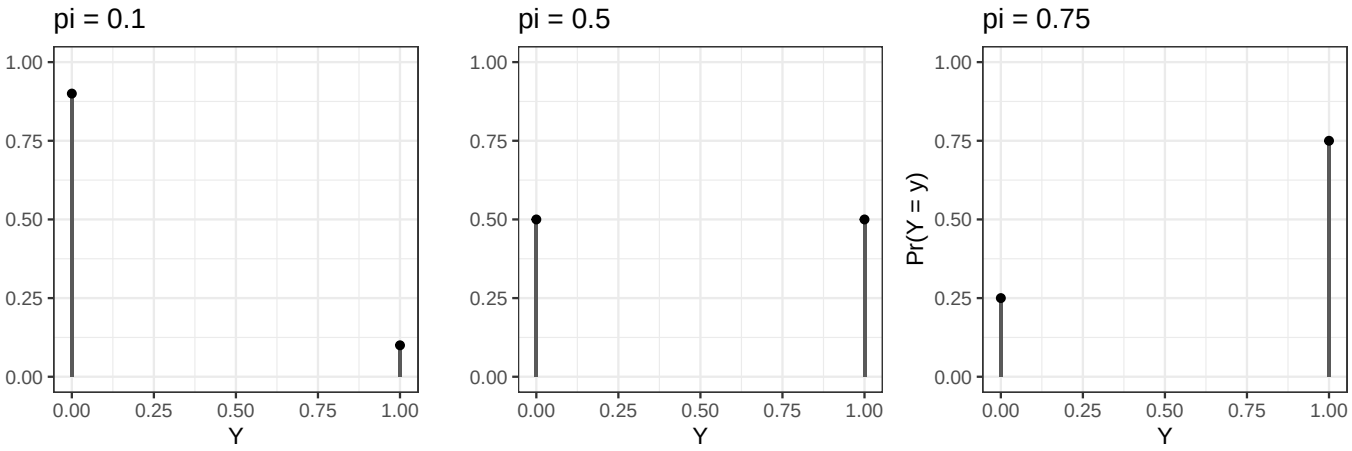
π probability of yes/"success"/$y = 1$
$\omega = \frac{\pi}{1-\pi}$ odds of yes/"success"/$y = 1$

- Suppose we're interested in lung cancer incidence (yes/no) in two groups smokers (1) and non-smokers (2)...

Model for Binary Response Variable

$$Y \sim Bernoulli(\pi)$$

- $Pr[Y = y] = \pi^y(1-\pi)^y$ for $y \in \{0, 1\}$
- $Mean\{Y\} = E[Y] = \pi$
- $Var\{Y\} = E[(Y-\pi)^2] = \pi(1-\pi)$
- How is this binary response fundamentally different from a continuous response ($Y \sim N(\mu, \sigma^2)$)?

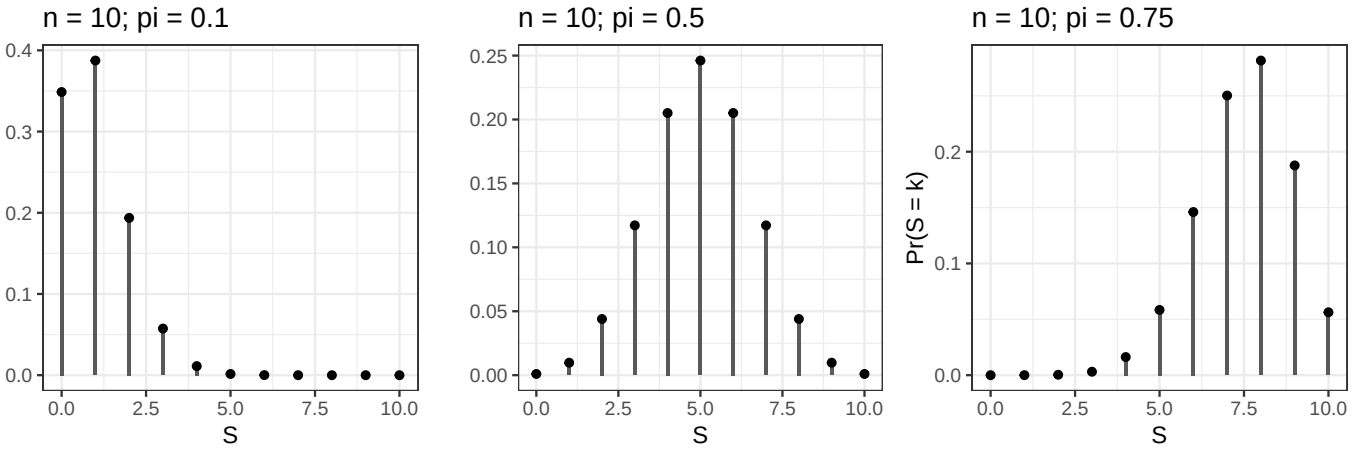


Binomial distribution

Let $\{Y_1, Y_2, ..., Y_n\}$ be RS from population of binary response variables with probability of yes equal to π

- $S = Y_1 + Y_2 + ... + Y_n$ is the total number Y = "yes" (1)
- $S \sim Binomial(n, \pi)$
- $Pr[S = k] = \binom{n}{k}\pi^k(1-\pi)^{n-k}$ for $S \in \{0, 1, 2, 3, ..., n\}$
- $Mean\{S\} = E[S] = n\pi$
- $Var\{S\} = E[(S - n\pi)^2] = n\pi(1-\pi)$
- When # expected "yes" $n\pi$ and # expected "no" $n\pi(1-\pi)$ are moderately large, binomial probabilities are approximated by a normal distribution!

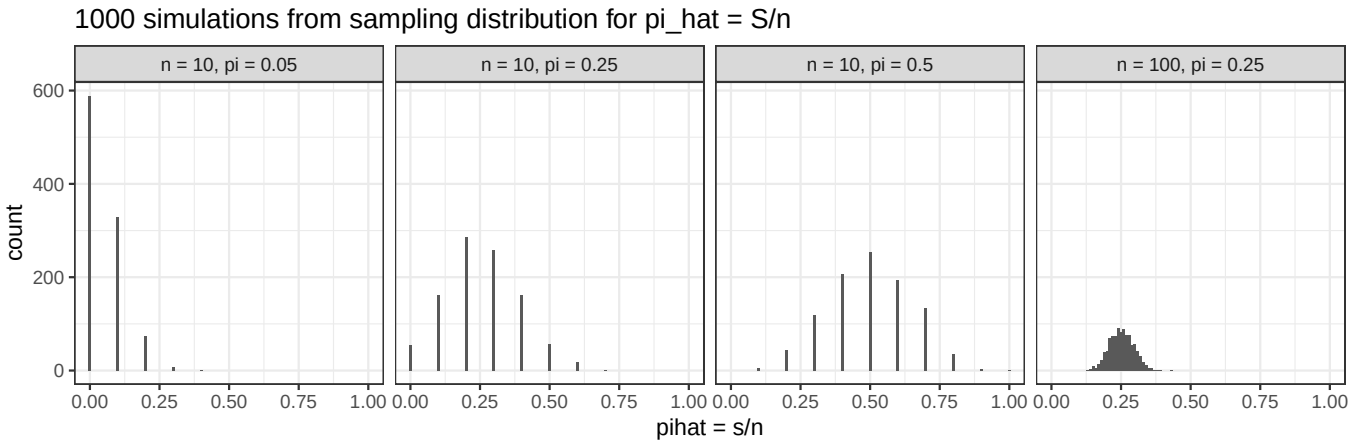
Visually



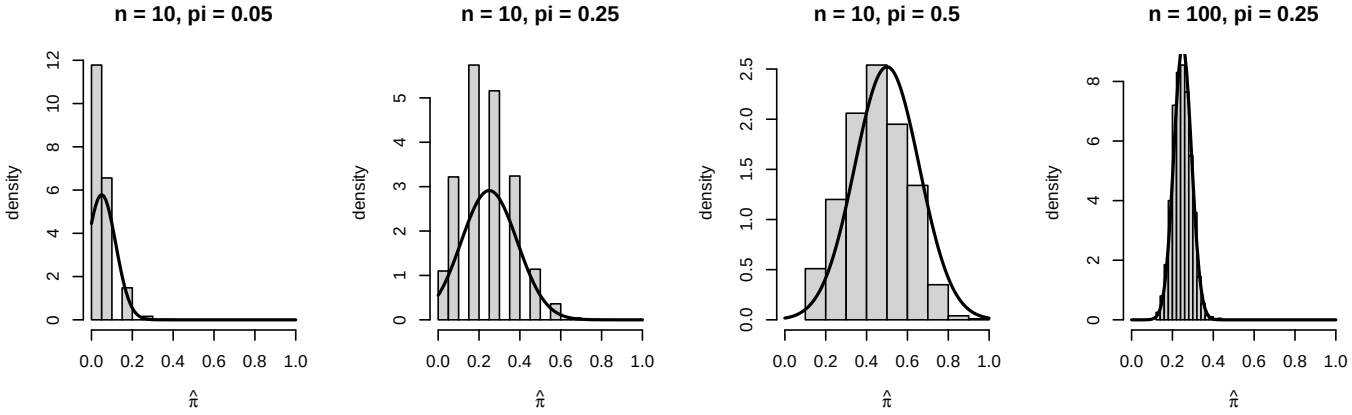
Observe RS of binary response data $\{Y_1, Y_2, ..., Y_n\}$, and want to estimate π

- Point estimate: $\hat{\pi} = S/n$
- Mean$\{\hat{\pi}\} = \pi$
- Var$\{\hat{\pi}\} = \pi(1-\pi)/n$
- If n is large enough, the sampling distribution of $\hat{\pi}$ is approximately normal!
  - Use $n\pi_0$ and $n(1-\pi_0)$ to assess if Normality is reasonable

Visually



Compared to Normal Approximation



Observe RS $\{Y_1, Y_2, ..., Y_{n_1}\}$ from group 1 and RS from $\{Y_1, Y_2, ..., Y_{n_2}\}$ group 2 and want to estimate $\pi_1 - \pi_2$

- Point estimate: $\hat{\pi}_1 - \hat{\pi}_2 = S_1/n_1 - S_2/n_2$
- Mean$\{\hat{\pi}_1 - \hat{\pi}_2\} = \pi_1 - \pi_2$
- Mean$\{\hat{\pi}_1 - \hat{\pi}_2\} = \frac{\pi_1(1-\pi_1)}{n_1} - \frac{\pi_2(1-\pi_2)}{n_2}$
- If n_1 and n_2 are large, the sampling distribution of $\hat{\pi}_1 - \hat{\pi}_2$ is approximately normal!
  - Use smaller of the sample sizes and a combined prob of yes to check that expected # of yes and expected number of no are "large enough"

Ch 18 has much more...

- Formulas for CIs and hypothesis tests for one mean and comparing two means are provided
- Analogous to X11 methods for inference for one mean or a difference in two groups means
- For comparing proportions we use Z-tests (Normal) rather than T-tests. Why?

Differences can be misleading

What values of π_1 and π_2 make the following true?

$$\pi_1 - \pi_2 = 0.05$$

⚠ Practical implication of $\pi_1 - \pi_2$ is non-constant!

Depends on the actual values of π_1 and π_2.

- When π_1 and π_2 are close to 0 or 1, practical implications tend to be much larger than when the parameter values are close to 0.5

So what do we do?

- We can compare odds across groups instead of differences in probabilities

Properties of odds

1. $\omega \in [0, \infty)$
2. $\pi = 0.5 \implies \omega = 1$
3. If odds of "yes" $= \omega$, odds of "no" $= \frac{1}{\omega}$
4. If odds of "yes" $= \omega$, probability of "yes" $\pi = \frac{\omega}{1+\omega}$

Practice interpreting odds

$\omega_1 = 5 \implies$

$\omega_1 = 0.5 \implies$

$\omega_1 = 0.8 \implies$

Estimating π and ω

	Y = *yes*	Y = *no*	
X = Group 1	$n_{1,yes}$	$n_{1,no}$	$n_1 = R_1$
X = Group 2	$n_{2,yes}$	$n_{2,no}$	$n_2 = R_2$
	$n_{yes} = C_1$	$n_{no} = C_2$	$n = T$

$\hat{\pi}$ estimates π $\hat{\pi} =$

$\hat{\omega}$ estimates ω $\hat{\omega} =$

Odds Ratio

$\phi = \omega_1/\omega_2$ is typically used to compare probabilities across groups

1. ϕ tends to remain more nearly constant (than $\pi_1 - \pi_2$) over levels of confounding variables
2. ϕ is only valid measure for *retrospective studies*
3. ϕ extends to logistic regression! (Ch 20/21)

$\hat{\phi}$ **estimates** ϕ $\hat{\phi} =$

Stat theory in a nutshell

If $\widehat{\omega_1}$ and $\widehat{\omega_2}$ are computed from independent samples, then the *sampling distribution of the log-odds* $log(\widehat{\omega_1}/\widehat{\omega_2})$...

1. Has approximate center equal to $log(\omega_1/\omega_2)$
2. Has approximate variance equal to $\frac{1}{n_1\pi_1(1-\pi_1)+n_2\pi_2(1-\pi_2)}$
3. If both n_1 and n_2 are large, $log(\widehat{\omega_1}/\widehat{\omega_2}) \dot{\sim} N(mean(1), var(2))$

Statistical inference for large n_1, n_2 (CIs)

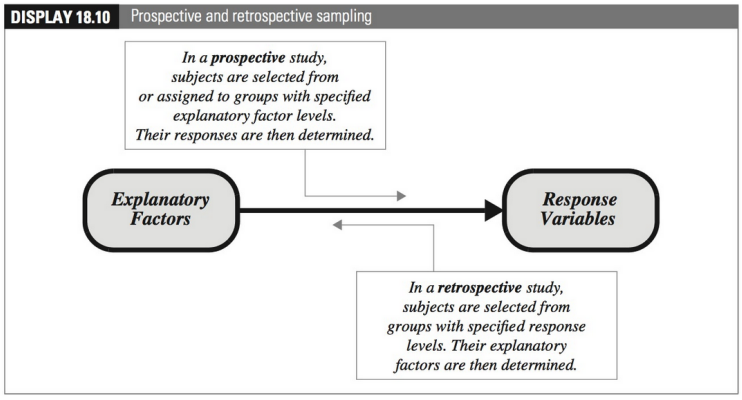
Statistical inference for large n_1, n_2 (tests)

Back to smoking and lung cancer

Interested in lung cancer incidence (yes/no) in two groups smokers (1) and non-smokers (2).

- Can we use a randomized design to assign smoking to participants?
- Can we identify groups of smokers and non-smokers within the same population and see if they develop lung cancer?
- More practical to sample participants with lung cancer and those without from the same population and ask them if they are a smoker.

Retrospective vs. prospective studies



ϕ for retrospective studies

DISPLAY 18.11 Illustration that the odds ratio is the same regardless of which factor is thought of as the response, with a hypothetical population

	Lung cancer	No cancer
Smokers	1,000	2,000,000
Nonsmokers	4,000	16,000,000

$$\frac{\text{Odds of cancer among smokers}}{\text{Odds of cancer among nonsmokers}} = \frac{1{,}000/2{,}000{,}000}{4{,}000/16{,}000{,}000} = 2$$

$$\frac{\text{Odds that a cancer victim was a smoker}}{\text{Odds that a person without cancer was a smoker}} = \frac{1{,}000/4{,}000}{2{,}000{,}000/16{,}000{,}000} = 2$$

Concept diagram for case study

Create 95% CI for log(ϕ)

	Cancer	Control	Totals
Smokers	83	72	155
Nonsmokers	3	14	17
Totals	89	86	172

Point Estimates

```
# sample prob cancer in smoking group
(phat1 <- 83/155)
```

```
[1] 0.5354839
```

```
# sample odds
(omhat1 <- phat1/(1-phat1))
```

```
[1] 1.152778
```

```
# sample prob cancer in non-smoking group
(phat2 <- 3/17)
```

```
[1] 0.1764706
```

```
# sample odds
(omhat2 <- phat2/(1-phat2))
```

```
[1] 0.2142857
```

```
# sample odds ratio
(phi_hat <- omhat1/omhat2)
```

```
[1] 5.37963
```

95% CI for $log(\phi)$

```
# 95% CI for log(phi): log_phihat +/- 1.96*SE(log(phihat))
lphi_hat <- log(phi_hat)
lphi_hat
```

```
[1] 1.68262
```

```
se_lphi_hat <- sqrt(1/(155*phat1*(1-phat1)) + 1/(17*phat2*(1-phat2)))
se_lphi_hat
```

```
[1] 0.6562766
```

```r
ci_log_odds <- lphi_hat + c(-1,1)*qnorm(p = 0.975)*se_lphi_hat
ci_log_odds
```

```
[1] 0.396341 2.968898
```

```r
exp(ci_log_odds)
```

```
[1]  1.486376 19.470452
```

Unpack and critique the Slueth summary

Example—Smoking and Lung Cancer

From the data described in Example 18.1.3 the estimated odds ratio is 5.38, the natural log of the estimated odds ratio is 1.68, and the standard error of the estimated log odds ratio is 0.656 (confidence interval version). An approximate 95% confidence interval for the log odds ratio is 0.396 to 2.969. Therefore, the odds of lung cancer for smokers are estimated to be 5.38 times the odds of lung cancer for nonsmokers. An approximate 95% confidence interval for the odds ratio is 1.5 to 19.4. As usual, no causal interpretation can be drawn from the statistical analysis of observational data, although the data are consistent with the theory that smoking causes lung cancer.

Take Aways

- Comparing odds is more appropriate when the population proportions are near 0 or 1
- Inferential methods are based on normality assumptions and require moderately large sample sizes for each group under study!
- Need larger n_1 and n_2 for normal approx to be reasonable when population proportions are near 0 or 1

Ch 19 higlights

Methods for tests of *homogeneity* and *independence* between two categorical factors (extension to more than 2 levels in both X and Y)

- Sampling scheme dictates interpretation and type of test

Fisher's Exact test $H_0 : \pi_1 - \pi_2 = 0$

Non-parametric randomization test that works for 2×2 tables

- No normality assumptions
- Provides null distribution for testing H_0
- Need to use another procedure for CI (bootstrap re-sampling)
- Sometimes requires extensive computation (for large samples)
- Can be approximated with Pearson's χ^2 test

For larger tables...

The *response variable is categorical* and there is interest in comparing the proportions within each level of the response across the levels of a *categorical explanatory variable.*

- Two potential hypotheses *homogeneity* and *independence*
- Study design determines which is most appropriate

Visually

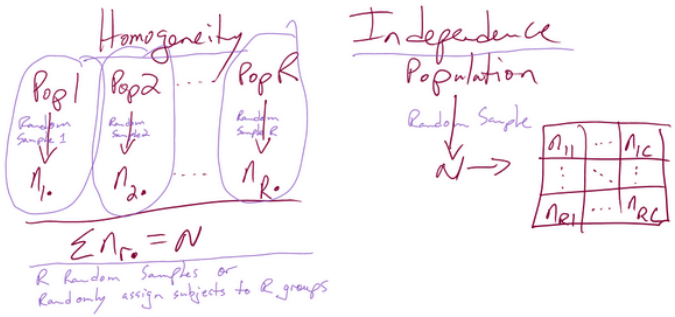


Figure 5.3: Diagram of the scenarios involved in Homogeneity and Independence tests. Homogeneity testing involves R random samples or subjects assigned to R groups. Independence testing involves a single random sample and measurements on two categorical variables.

Figure 1: Dr. Mark Greenwood's sketch from 337 text

Hypotheses for *Homogeneity*

- H_0: The population distributions of the response is the same across the groups defined by the explanatory variable
- H_a: **At least one** of the population distributions of the response is not the same across the groups defined by the explanatory variable.

Hypotheses for *Independence*

- H_0: There is no relationship between X and Y in the population.
- H_a: There is some relationship (an association) between X and Y in the population.

Homogeneity or independence?

Suppose you want to...

- Investigate the association between travelers who think it is rude to recline your airplane seat (no, somewhat, yes) and the frequency with which they recline their seat (never, once in a while, about half the time, usually, always)
- Compare the distribution of a 5-species assemblage of bats (MYCI, MYLU, EPFU, LANO, LACI) across different regions in the United States (SW, NW, Midwest, South, NE, SE). Sampling efforts occur in each of the regions.

Pearson's χ^2 test

- Allows for more than two categories in the "response"
- Mechanics are the same for tests of *homogeneity* and *independence*
  - $X^2 = \sum_{r=1}^{R}\sum_{c=1}^{C} \frac{(Observed_{rc} - Expected_{rc})^2}{Expected}$
  - $X^2 \dot{\sim} \chi^2_{(r-1)\times(c-1)}$ under H_0
- Drawback: not very informative, and no CI available!
- Are the basis for omnibus "goodness of fit" tests

Summary

- Book outlines different types of sampling for "homogeneity" and "independence" tests, and connections to probability distributions
- Book outlines tests (no CIs for these methods)
- We will focus on binary and Binomial logistic regression (Ch 20, 21)